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C-type natriuretic peptide improves maternally aged oocytes quality by inhibiting excessive PINK1/Parkin-mediated mitophagy

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Abstract

The overall oocyte quality declines with ageing, and this effect is strongly associated with a higher reactive oxygen species (ROS) level and the resultant oxidative damage. C-type natriuretic peptide (CNP) is a well-characterized physiological meiotic inhibitor that has been successfully used to improve immature oocyte quality during in vitro maturation (IVM). However, the underlying roles of CNP in maternally aged oocytes have not been reported. Here, we found that the age-related reduction in the serum CNP concentration was highly correlated with decreased oocyte quality. Treatment with exogenous CNP promoted follicle growth and ovulation in aged mice and enhanced meiotic competency and fertilization ability. Interestingly, the cytoplasmic maturation of aged oocytes was thoroughly improved by CNP treatment, as assessed by spindle/chromosome morphology and redistribution of organelles (mitochondria, the endoplasmic reticulum [ER], cortical granules [CGs], and the Golgi apparatus). CNP treatment also ameliorated DNA damage and apoptosis caused by ROS accumulation in aged oocytes. Importantly, oocyte RNA-seq revealed that the beneficial effect of CNP on aged oocytes was mediated by restoration of mitochondrial oxidative phosphorylation, eliminating excessive mitophagy. CNP reversed the defective phenotypes in aged oocytes by alleviating oxidative damage and suppressing excessive PINK1/Parkin-mediated mitophagy. Mechanistically, CNP functioned as a cAMP/PKA pathway modulator to decrease PINK1 stability and inhibit Parkin recruitment. In summary, our results demonstrated that CNP supplementation constitutes an alternative therapeutic approach for advanced maternal age-related oocyte deterioration and may improve the overall success rates of clinically assisted reproduction in older women.

eLife assessment

This study presents **valuable** findings on the impact of C-natriuretic peptide (CNP) treatment in vivo on the fertility of aged mice. **Solid** data indicate CNP induces the cAMP-PKA pathway, causing reduced recruitment of Parkin protein to mitochondria in oocytes, resulting in reduced mitophagy which may be significant for increased mitochondrial bioenergetics and improved cytoplasmic and nuclear maturation. The authors make additional claims regarding the mechanisms by which CNP impacts oocyte quality in vivo for which the evidence is **incomplete**. This work will be of interest to reproductive biologists and clinical infertility specialists.

Introduction

During growth, oocytes gradually acquire the capacity to resume meiosis, complete maturation, undergo successful fertilization and achieve subsequent embryo developmental competence (Gandolfi & Gandolfi, 2001). However, ovarian involution precedes that of any other organ in female mammals, and in humans, the oocyte fertilization rate decreases rapidly after 35 years of age. Indeed, infertility associated with a decline in oocyte quality with increasing maternal age is a significant challenge.

Recently, with the development of cultural and social trends, many women have delayed childbearing, and ovarian senescence has become a public health problem (Bartimaeus, Obi, Igwe, & Nwachuku, 2020; Broekmans, Soules, & Fauser, 2009). Ovarian ageing is accompanied by abnormalities in organelle distribution, morphology and functions, leading to inadequate oocyte growth, maturation, fertilization and subsequent embryo development (Reader, Stanton, & Juengel, 2017). Consequently, assisted reproductive technologies such as in vitro maturation (IVM) and in vitro fertilization (IVF) have become promising options for infertility treatment (Chang, Song, Lee, Lee, & Yoon, 2014). However, further studies are needed to improve the subsequent developmental competence of maternally aged oocytes.

Oocyte maturation has two steps: nuclear maturation, which mainly involves germinal vesicle breakdown (GVBD) and chromosomal segregation, and cytoplasmic maturation, which involves redistribution of organelles (mitochondria, cortical granules [CGs], and the endoplasmic reticulum [ER]), changes in the intracellular ATP and antioxidant contents, and the accumulation of fertilization-related transcripts and proteins (McClatchie et al., 2017; Watson, 2007). The quality and developmental potential of aged oocytes are lower than those of oocytes derived from young females, primarily because aged oocytes exhibit negative consequences of cytoplasmic maturation, such as abnormal mitochondria and an aberrant CG distribution (Miao, Cui, Gao, Rui, & Xiong, 2020), deteriorated organelle and antioxidant system function and increased reactive oxygen species (ROS) levels (T. Zhang et al., 2019). Excessive ROS generation leads to destructive effects on cellular components (Zarkovic, 2020). Notably, recent studies have indicated that increased oxidative damage is closely correlated with the occurrence of mitochondrial damage and mitophagy (Jiang et al., 2021), which is accompanied by blockade of oocyte meiosis (Jin et al., 2022; Q. Shen, Liu, Li, & Zhang, 2021).

The endogenous C-type natriuretic peptide (CNP) produced by follicular mural granulosa cells as a ligand of natriuretic peptide receptor 2 (NPR2), which is expressed primarily in cumulus cells, plays a crucial role in maintaining meiotic arrest (Meijia Zhang, Su, Sugiura,

Xia, & Eppig, 2010). Recent studies have suggested that CNP, an inhibitor of oocyte maturation, provides adequate time for cytoplasmic maturation, offering a new strategy to optimize the synchronization of nuclear and cytoplasmic maturation and improve the quality of immature oocytes in vitro (Wei et al., 2017). Moreover, CNP has been reported to enhance the antioxidant defence ability and developmental competence of oocytes in vitro (Zhenwei & Xianhua, 2019). Therefore, CNP may constitute a new alternative means to enhance antioxidant system function and protect against oxidative damage by eliminating excess ROS in aged oocytes. Although CNP has been suggested to contribute to improving the maturation and subsequent development of immature mouse oocytes in vitro, the effect of CNP on maternally aged oocytes remains to be determined.

In this study, we investigated the redistribution and function of organelles (mitochondria, CGs, and the ER), the ATP content and the intracellular GSH level in CNP-treated maternally aged oocytes. The results showed that CNP improves the cytoplasmic maturation and developmental competence of maternally aged oocytes by optimizing organelle distribution and function and inhibiting PINK1/Parkin-mediated mitophagy. The findings of this study will contribute to understanding the mechanism of CNP in increasing the fertilization capacity and developmental ability of aged oocytes.

Results

CNP supplementation improves the quality of aged oocytes

To explore the effect of CNP on oocyte quality in aged mice, we first investigated whether intraperitoneal injection of CNP can affect oocyte quality. Young and aged mice were hormonally superovulated after 14 days of consecutive PBS or CNP treatment (Figure 1A, B). Then, the mice were sacrificed, and the oocytes were removed for measurement of the CNP content. As expected, the endogenous CNP content in serum from aged mice was markedly lower than that in serum from young mice (Figure 1C). In contrast, administration of CNP to aged mice significantly elevated the CNP content in serum (Figure 1C). To further determine whether the elevated CNP content in serum can improve oocyte quality, we evaluated the number, first polar body (PB1) extrusion rate and fragmentation rate. As the mice aged, the number of ovulations and the PB1 extrusion rate decreased significantly, but the incidence of fragmented oocytes increased dramatically (Figure 1D-G). Conversely, CNP supplementation apparently ameliorated the ageing-induced defects in the number and morphology of the ovulated oocytes (Figure 1D-G). In addition, assessment of follicle development in the ovary sections by HE staining showed severe deterioration of follicles at different developmental stages in aged mice; however, CNP supplementation significantly increased the number of secondary follicles and antral follicles (Figure 1H and I). Young, untreated aged and CNP-treated aged mice were naturally mated with 12-week-old male mice, and consistent with the increased number of ovulated oocytes with a normal morphology, the litter size of aged mice was also increased by CNP administration (Figure 1J).

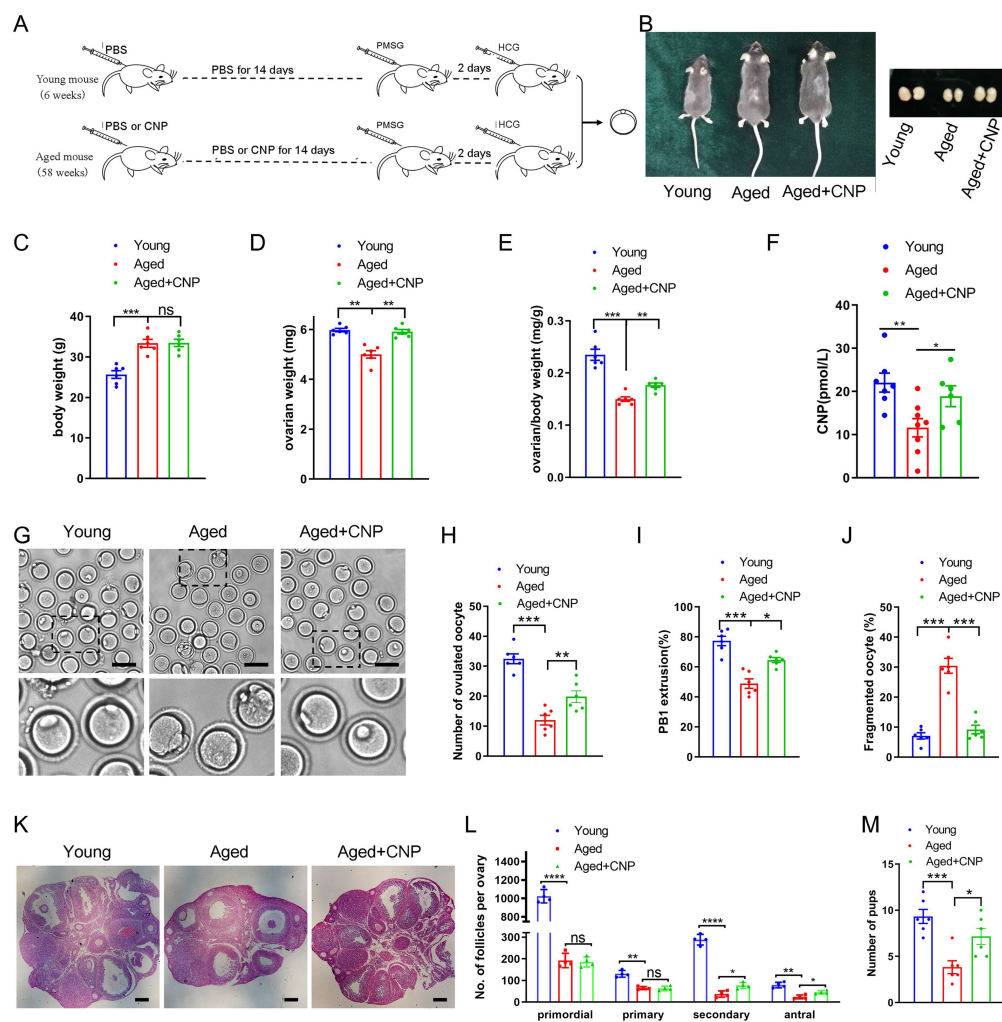


Figure 1.

Effects of CNP supplementation on the oocyte quality and female fertility in aged mice.

(A) A timeline diagram of CNP administration and superovulation. (B) Representative images of young, aged and CNP administration aged (Aged + CNP) mice as well as their ovaries. (C) Body weights of young, aged and aged + CNP aged mice. (D) Ovarian weights of young, aged and aged + CNP mice. (E) Ratios of ovarian weight to body weight for each group of mice. (F) Serum CNP concentrations were measured in young, aged, and aged + CNP mice. (G) Representative images of the oocyte polar body extrusion in young, aged, and aged + CNP mice. Scale bar: 100 μ m. (H) Ovulated oocytes were counted in young, aged, and aged + CNP mice. (I) Rate of polar body extrusion in young, aged, and aged + CNP mice. (J) The rate of fragmented oocytes was recorded in young, aged, and aged + CNP mice. (K) Representative images of ovarian sections from young, aged, and aged + CNP mice. Scale bars: 100 μ m. (L) Follicles at different developmental stages were counted in young, aged, and aged + CNP ovaries. (M) Average litter size of mated mice was assessed by mating with two-months old male mice.

To investigate the effects of CNP on in vitro maturation (IVM) of aged mouse cumulus-oocyte complexes (COCs), we first examined the PB1 extrusion rate of COCs pretreated with 10 nM CNP to maintain meiotic arrest for 24 h (pre-IVM) and then matured in vitro for 16 h (a two-

step culture system; Figure S1 A). In the control (conventional in vitro maturation) group, only $35.36 \pm 2.74\%$ of the oocytes exhibited PB1 extrusion. After temporary meiotic arrest induced by treatment with 10 nM CNP, the maturation rate increased to $71.12 \pm 3.02\%$ ($n=104$), significantly higher than that in the control group ($P<0.01$) (Figure S1 B and C). The spindle morphology and chromosome alignment in in vitro-matured oocytes were also evaluated. The percentage of oocytes with abnormal spindle-chromosome complexes was significantly decreased in the group with 10 nM CNP-induced temporary meiotic arrest (Figure S1 D and E). Collectively, these results indicate that CNP administration increased the serum CNP content, restored the number and morphology of aged oocytes and improved the fertility of aged female mice.

CNP supplementation restores cytoplasmic maturation events in maternally aged mouse oocytes

Pregnancy failure and foetal miscarriage increase with maternal age and, importantly, are associated with oocyte aneuploidy and spindle/chromosomal abnormalities (Ma et al., 2020). Therefore, we determined the rate of spindle/chromosomal abnormalities in oocytes of young, untreated aged and CNP-treated aged mice by immunofluorescence staining and found that CNP treatment greatly improved the spindle/chromosomal abnormalities in aged mice (Figure 2A, B). To determine whether, in addition to affecting spindles/chromosomes, CNP supplementation affects other organelles during the maturation of aged oocytes, we examined the distribution of the Golgi apparatus, endoplasmic reticulum (ER) and cortical granules (CGs) in oocytes from young, untreated aged and CNP-treated aged mice. The Golgi-Tracker results showed that in aged mouse oocytes, the Golgi apparatus was distributed in agglutinated and clustered patterns, and CNP supplementation significantly reduced the rate of abnormal Golgi distribution (Figure 2 C, D, Figure S2A, B). Since the ER plays an essential role in Ca^{2+} signal-mediated oocyte fertilization and subsequent embryonic development (Miyazaki & Ito, 2006), we then examined the distribution pattern of the ER in oocytes. As shown in Figure 2E, the ER was accumulated at the chromosome periphery and was evenly distributed in the cytoplasm; however, the ER abnormally agglomerated in the cytoplasm and the chromosome periphery in a disorganized pattern in aged oocytes (Figure 2E, Figure S3A). Statistical analysis showed that the rate of abnormal ER distribution was significantly decreased in CNP-supplemented oocytes (Figure 2F, Figure S3B). The distribution of CGs is one of the most important indicators of oocyte cytoplasmic maturation and is related to the blockade of polyspermy following fertilization. We assessed whether CNP supplementation affects the distribution dynamics of CGs in aged oocytes. Lens culinaris agglutinin (LCA)-FITC staining showed that in young oocytes, CGs were distributed evenly in the oocyte subcortical region, leaving a CG-free domain (CGFD) near chromosomes (Figure 2G). However, maternally aged oocytes showed an abnormal CG distribution, including increased migration of CGs towards the oocyte chromosomes or oocyte subcortical region, without leaving a CGFD (Figure 2G, H; Figure S4 A). Consistent with this finding, statistical analysis of the fluorescence intensity of CG signals in aged oocytes showed a significant reduction compared with that in young oocytes, and CNP supplementation rescued the mislocalization and decrease in the number of oocyte CGs (Figure 2 H, I, Figure S4 B, C). Taken together, these data imply that CNP is a potent agent for rescuing cytoplasmic maturation events in maternally aged mouse oocytes.

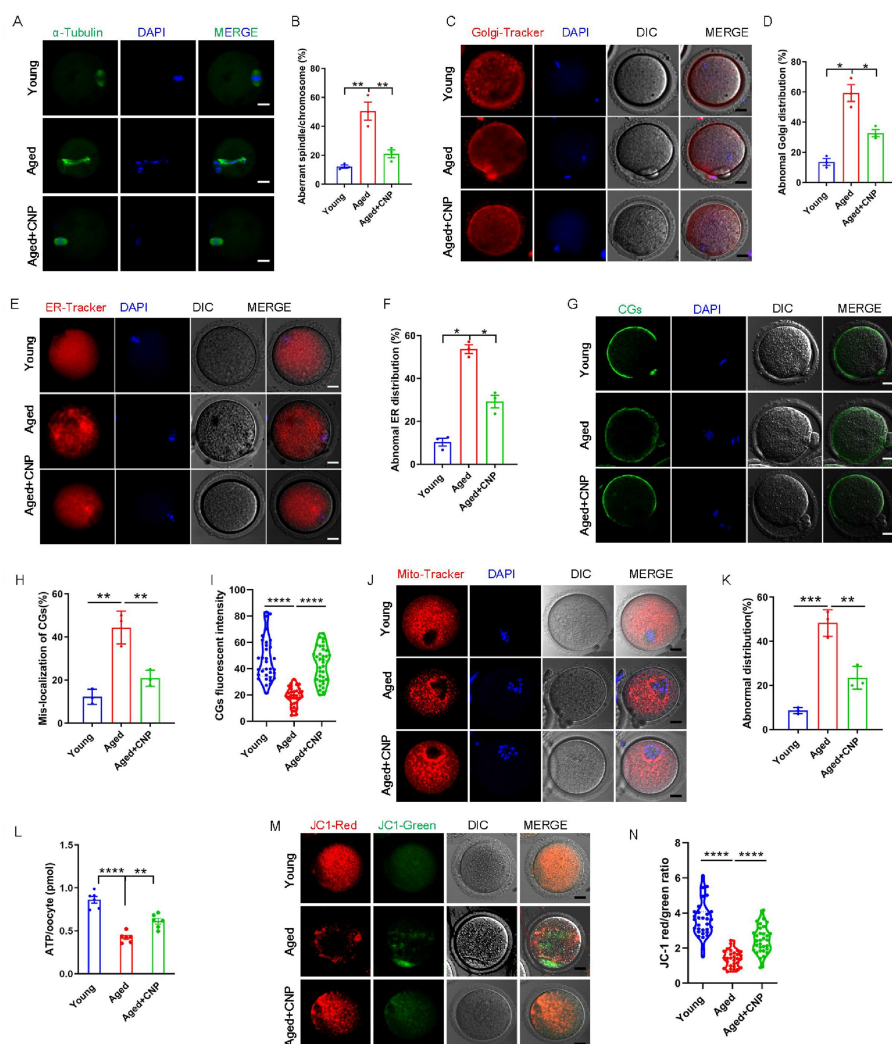


Figure 2.

CNP supplementation recovers cytoplasmic maturation events of maternally aged mouse oocytes.

(A) Representative images of the spindle morphology and chromosome alignment at metaphase II in young, aged, and aged + CNP mice. Scale bar, 10 μ m. (B) The rate of aberrant spindles at metaphase II was recorded in young, aged, and aged + CNP mice. (C) Representative images of the Golgi apparatus distribution at metaphase II in young, aged, and aged + CNP mice. Scale bar, 10 μ m. (D) The rate of aberrant Golgi apparatus distribution was recorded in young, aged, and aged + CNP mice. (E) Representative images of the endoplasmic reticulum distribution at metaphase II in young, aged, and aged + CNP mice. Scale bar, 10 μ m. (F) The rate of aberrant endoplasmic reticulum distribution was recorded in young, aged, and aged + CNP mice. (G) Representative images of the cortical granules (CGs)

distribution in young, aged, and aged + CNP mice. Scale bar, 10 μ m. (H) The rate of mislocalized CGs was recorded in the young, aged, and aged + CNP mice. (I) The fluorescence intensity of CG signals was measured in the young, aged, and aged + CNP mice oocyte. (J) Representative images of mitochondrial distribution in the young, aged, and aged + CNP mice oocytes stained with MitoTracker Red. Scale bar, 10 μ m. (K) The abnormal rate of mitochondrial distribution was recorded in the young, aged, and aged + CNP mice oocytes. (L) ATP levels were measured in the young, aged, and aged + CNP mice. (M) Mitochondrial membrane potential ($\Delta\Psi$ m) was detected by JC-1 staining in the young, aged, and aged + CNP mice oocytes. Scale bar, 10 μ m. (N) The ratio of red to green fluorescence intensity was calculated in the young, aged, and aged + CNP mice oocytes.

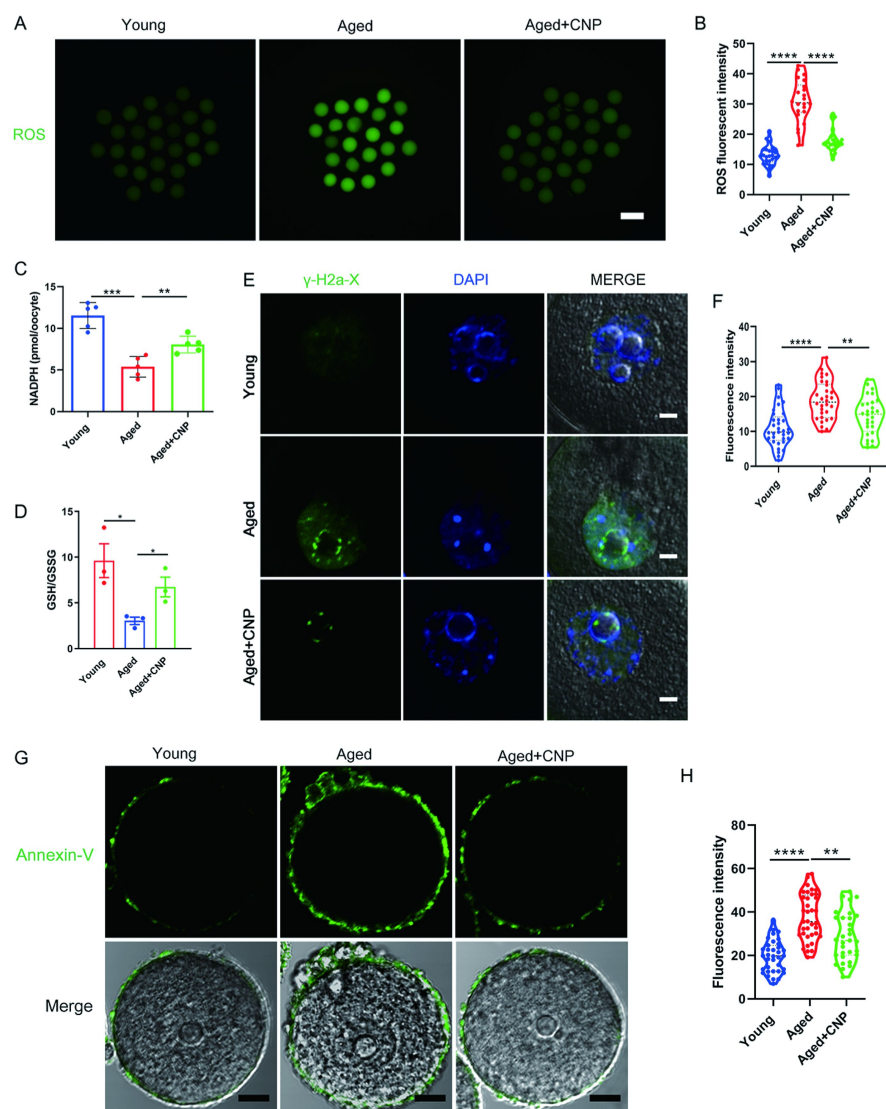
CNP supplementation restores mitochondrial distribution and function in aged oocytes

To verify the effect of CNP supplementation on the mitochondrial distribution pattern and function in aged oocytes, we performed MitoTracker staining. In young oocytes, mitochondria exhibited a homogeneous distribution in the cytoplasm and accumulated at the periphery of chromosomes (Figure 2J). However, in aged oocytes, most mitochondria were aggregated in the cytoplasm and partially or completely failed to accumulate around

chromosomes (Figure 2JA, Figure S5A). Statistically, more than 40% of mitochondria in aged oocytes exhibited a mislocalized distribution pattern, and CNP supplementation significantly reduced the abnormal distribution rate (Figure 2K, Figure S5B). We then analysed mitochondrial function by measuring the ATP content in oocytes from young, untreated aged, and CNP-treated aged mice. The ATP content in oocytes from aged mice was considerably lower than that in oocytes from young mice but was restored following CNP supplementation (Figure 2L, Figure S5C). We also tested the mitochondrial membrane potential, which has been shown to be the driving force of mitochondrial ATP synthesis, by staining with the potentiometric dye JC-1 (Figure 2M, Figure S5D). The mitochondrial membrane potential was lower in oocytes from aged mice than in oocytes from young mice but was restored in oocytes from CNP-supplemented aged mice (Figure 2M, N, Figure S5D, S5E). Overall, these observations suggest that CNP supplementation improved ageing-induced mitochondrial dysfunction in oocytes.

CNP supplementation eliminates excessive ROS and attenuates DNA damage and apoptosis in aged oocytes

We proposed that mitochondrial dysfunction induces ROS imbalance and oxidative stress in aged oocytes. To test this hypothesis, we carried out dichlorofluorescein (DCFH) staining to measure ROS levels in each group of oocytes (Figure 3A). Quantitative analysis of the fluorescence intensity showed that ROS signals were markedly enhanced in aged oocytes compared with young oocytes (Figure 3B). Conversely, CNP supplementation effectively reduced the ROS accumulation observed in aged oocytes (Figure 3A and 3B, Figure S6A, S6B). In addition to being caused by ROS accumulation, age-associated oxidative stress damage can be caused by reduced antioxidant defence system function. We therefore investigated whether CNP contributes to improving the antioxidant defence ability in aged oocytes. Quantification of the nicotinamide adenine dinucleotide phosphate (NADPH) levels and the ratio of reduced to oxidized glutathione (GSH/GSSG ratio) in oocytes showed that NADPH levels and the GSH/GSSG ratio were decreased in oocytes from aged mice compared with those from young mice and that CNP treatment significantly increased NADPH levels and the GSH/GSSG ratio in oocytes from aged animals (Figure 3C and 3D). Because a high level of ROS not only results in the accumulation of DNA damage but also causes oocyte apoptosis (Miao et al., 2020), we next evaluated DNA damage and apoptosis in oocytes by γ -H2A.X and Annexin-V staining, respectively. As expected, higher signals indicating DNA damage and apoptosis were observed in aged oocytes than in young oocytes, and these increases were alleviated by supplementation with CNP (Figure 3E-3H). Taken together, these observations suggested that the rates of DNA damage and apoptosis are higher in aged oocytes, possibly because of maternal ageing-induced excessive accumulation of ROS. Notably, our results demonstrated that CNP supplementation exerts antioxidant activity, which is an effective strategy to ameliorate maternal ageing-induced DNA damage and apoptosis in oocytes.



CNP supplementation improves the fertilization ability and early embryo development of aged oocytes

Considering that oocyte fertilization and subsequent embryo developmental competence are profoundly affected by mitochondrial function, we then tested whether the oocyte fertilization capacity and normal development to the blastocyst stage are enhanced by CNP. The *in vitro* fertilization (IVF) results showed that aged oocytes had dramatically lower fertilization rates than young oocytes and that CNP supplementation effectively increased the fertilization rate of aged oocytes (Figure 4A, B). We further examined the subsequent developmental ability of the fertilized oocytes. As expected, CNP supplementation effectively increased the blastocyst formation rate of aged oocytes both *in vivo* (Figure 4A and 4C–4F) and *in vitro* (Figure S7). These results demonstrate that CNP increases the fertilization capacity and promotes subsequent embryonic development of oocytes from aged mice.

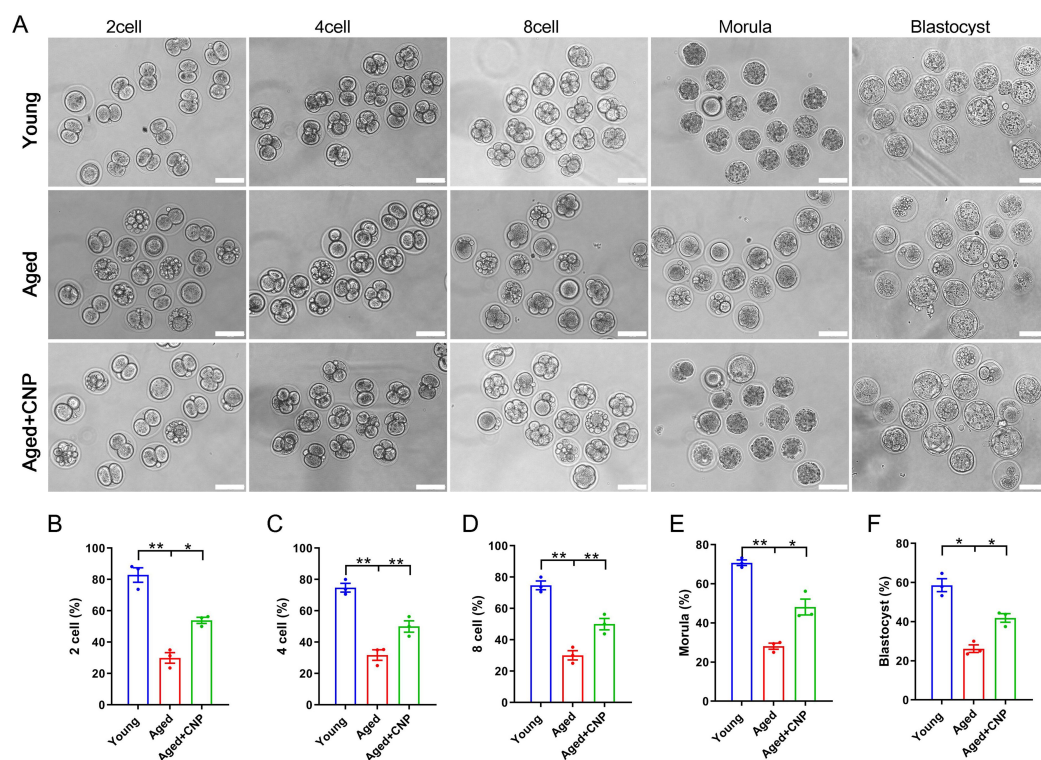


Figure 4.

Effects of CNP on the fertilization ability and embryonic development of aged oocytes.

(A) Representative images of early embryos developed from young, aged, and aged + CNP oocytes in vitro fertilization. Scale bar, 100 μ m. (B) The fertilization rate (2 cell embryos rate), (C) 4 cell embryos rate, (D) 8 cell embryos rate, (E) morula rate and (F) blastocyst formation rates were recorded in the young, aged, and aged + CNP groups. Data in (B)–(F) are presented as mean percentage (mean $\pm$ SEM) of at least three independent experiments.

Identification of target effectors of CNP in aged oocytes by single-cell transcriptome analysis

To verify the cellular and molecular mechanisms of CNP supplementation in improving oocyte quality in aged mice, we performed single-cell transcriptome analysis of GV oocytes derived from young, untreated aged and CNP-treated aged mice to identify potential target effectors. The relative expression of several randomly selected genes from each group was verified using quantitative real-time PCR (Figure S8A, B). As shown in the heatmap and volcano plot, the transcriptome profile of aged oocytes was significantly different from that of young oocytes, with 77 differentially expressed genes (DEGs) downregulated and 440 DEGs upregulated in aged oocytes identified through DEGseq2 analysis (Figure 5A–C). Furthermore, CNP supplementation resulted in downregulation of 584 genes and upregulation of 527 genes compared with aged oocytes. In particular, Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analysis showed that genes enriched in the ubiquitin-mediated proteolysis and mitophagy pathways were abnormally highly expressed in aged oocytes compared with young oocytes but that these expression levels were restored

to the baseline levels in CNP-supplemented aged oocytes (Figure 5D, E). In addition, oxidative phosphorylation and peroxisome proliferator-activated receptor (PPAR) signalling pathways were ranked at the top of the enrichment list in CNP-supplemented aged oocytes compared to untreated aged oocytes, consistent with our abovementioned observations that CNP supplementation improved mitochondrial function in aged oocytes. Many of the enriched KEGG enrichment pathways are highly related to mitophagy and mitochondrial function, which suggests that mitophagy should be strongly considered as a CNP effector in aged oocytes.

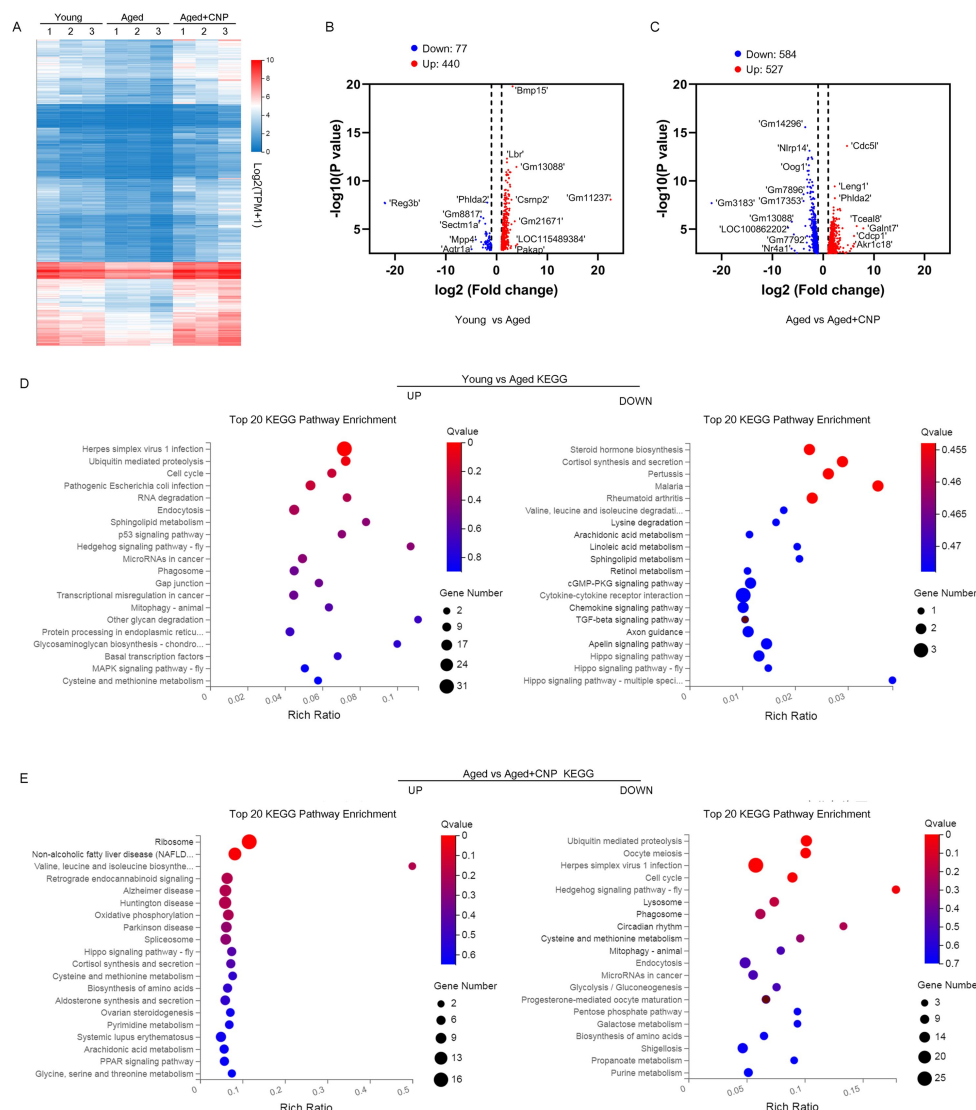


Figure 5.

Effect of CNP supplementation on transcriptome profiling of aged oocytes

(A) Heatmap illustration displaying gene expression of young, aged, and aged+ CNP oocytes. (B) Volcano plot showing differentially expressed genes (DEGs; downregulated, blue; upregulated, red) in Young vs Aged oocytes. Some highly DEGs are listed. (C) Volcano plot showing DEGs in Aged vs Aged + CNP oocytes. Some highly DEGs are listed. (D) KEGG enrichment analysis of upregulated and downregulated DEGs in Young vs Aged oocytes. (E) KEGG enrichment analysis of upregulated and downregulated DEGs in Aged vs Aged + CNP oocytes.

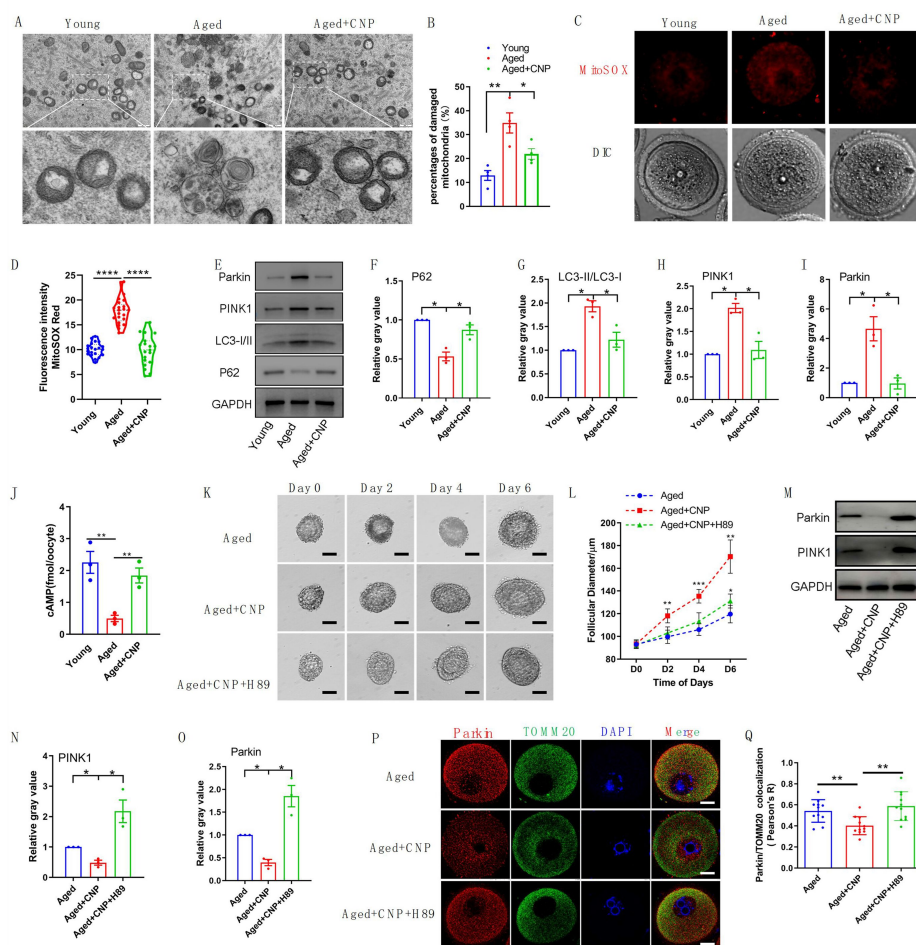
CNP supplementation attenuates oxidative damage by inhibiting mitophagy in aged oocytes

To verify the effect of CNP supplementation on mitophagy in aged oocytes, we first analysed mitochondrial structure in the oocytes of young, untreated aged and CNP-treated aged mice by transmission electron microscopy (Figure 6A). Quantitatively, mitochondrial damage, as evidenced by membrane rupture and a lack of electron density was significantly increased in aged oocytes but was ameliorated in CNP-supplemented aged oocytes (Figure 6B). Because oxidative stress has been implicated in triggering mitochondrial oxidative injury and mitophagy (Adhikari, Lee, Yuen, & Carroll, 2022; M. Shen et al., 2016), we next determined whether supplementation with CNP can eliminate excessive mitochondrial ROS (mtROS). As expected, supplementation with CNP substantially reduced the mtROS signals, as shown by MitoSOX staining and fluorescence intensity measurements (Figure 6C, D). We then evaluated degradation of the autophagy biomarker p62, the accumulation of LC3-II, the conversion of LC3-I to LC3-II, and the expression patterns of the mitophagy-related proteins PINK1 and Parkin (Figure 6E). Western blot analysis revealed that aged oocytes exhibited significant p62 degradation, LC3-II accumulation and marked increases in PINK1 and Parkin expression levels, whereas CNP supplementation abrogated these effects (Figure 6E–6I). Collectively, the above data indicate the inhibitory effect of CNP on oocyte mitophagy through the PINK1-parkin signalling pathway.

Figure 6.

Evaluation of CNP supplementation on mitophagy activity of aged oocytes.

(A) Representative images of mitochondria morphology and structure in young, aged, and aged + CNP oocytes by TEM. (B) Accumulation of mitochondria damage in young, aged, and aged + CNP oocytes. Under TEM images, percentages of damaged mitochondria per area (500 nm × 500 nm) were shown. At least 4 visions were chosen and mitochondria were counted by two individuals. (C) Representative images of mitochondria ROS stained with MitoSOX in young, aged, aged + CNP oocytes. Scale bar, 20 μm. (D) Fluorescence intensity of MitoSOX signals was measured in young aged, aged + CNP oocytes. (E) Western blots of P62, LC3-I/II, PINK1 and Parkin in young, aged, and aged + CNP oocytes. GAPDH was used as internal control. (F–I)



Relative gray value of proteins detected with western blots as compared with controls. (J) Oocyte cAMP concentrations were measured in young, aged, and aged + CNP mice. (K) Representative images at day 0, day 2, day 4 and day 6 of cultured preantral follicles with or without CNP or CNP+H89 treatment. Scale bar=50µm. (L) Diameters of preantral follicles with or without CNP or CNP+H89 treatment from day 0 to day 6. Six independent culture experiments were performed. (M) Western blots of PINK1 and Parkin in aged, aged + CNP and aged + CNP+H89 treated oocytes. GAPDH was used as internal control. (N-O) Relative gray value of proteins detected with western blots as compared with controls. (P) Double immunofluorescence staining of Parkin and TOMM20. The mitochondria outer membrane protein TOMM20 was performed to reveal the translocation of PRKN proteins on mitochondria. Red, PRKN; Green, TOMM20; Blue, DNA was labeled with Hoechst 33342. Bar: 20 µm. (Q) The colocalization of Parkin and TOMM20 in oocytes from aged, aged + CNP and aged + CNP+H89 treated mice were compared. Pearson's R shows the results of co-location analysis.

CNP downregulates Parkin recruitment and mitophagy via the cAMP/PKA pathway

How PINK1- and Parkin-mediated mitophagy is regulated by CNP in aged oocytes, however, requires further elucidation. The cAMP/PKA signalling pathway, which is dependent on the phosphorylation of mitochondrial proteins, has emerged as a direct means to regulate mitophagy and mitochondrial physiology (Amer & Hebert-Chatelain, 2018; Lobo et al., 2020). The concentrations of cAMP in GV oocytes derived from young, untreated aged and CNP-treated aged mice were determined. As shown in Fig. 6J, the cAMP concentration in aged oocytes was significantly lower than that in young oocytes, but administration of CNP resulted in a substantial increase in intraoocyte cAMP concentrations. This increase in cAMP significantly reduced mitochondrial recruitment of Parkin and mitophagy, which were dependent on PKA activity (Lobo et al., 2020). Next, we applied a PKA inhibitor, H89, to determine whether PKA is directly involved in CNP-mediated oocyte mitophagy. We isolated preantral follicles (80–100 µm diameter) from the ovaries of aged mice and treated them with 100 nM CNP or 100 nM CNP + 10 µM H89 during in vitro culture. Monitoring of follicle growth dynamics showed that treatment with 100 nM CNP significantly increased the follicle diameter (Figure 8B and C), whereas H89 treatment inhibited the effect of CNP on promoting preantral follicle growth (Figure 6 K and L). Western blot analysis revealed that CNP supplementation significantly decreased PINK1 and Parkin expression levels, but H89 treatment abrogated these expression changes (Figure 6 M-O). The cAMP-PKA pathway plays an important role in inhibiting Parkin recruitment to damaged mitochondria (Akabane et al., 2016). We therefore sought to determine whether PKA inhibition regulates Parkin recruitment. The effects of CNP on mitochondria were examined by double staining for Parkin and translocase of outer mitochondrial membrane 20 (TOMM20). CNP clearly inhibited the mitochondrial localization of Parkin, but inhibition of PKA with H89 resulted in Parkin translocation to mitochondria, as shown by the overlap of the two staining signals (Figure 6 P and Q). Collectively, these data suggested that the suppression of Parkin recruitment through the cAMP-PKA axis is an important mechanism underlying the protective effect of CNP against oxidative injury in maternally aged mouse oocytes.

Discussion

In mammals, the endogenous peptide CNP is expressed by endothelial cells in many tissues and has diverse physiological functions in mediating cardioprotective effects, bone growth, oocyte meiotic progression, and follicle growth and development (Bae et al., 2017; Moyes & Hobbs, 2019; Peake et al., 2014; Sato, Cheng, Kawamura, Takae, & Hsueh, 2012; Xi et al., 2019).

Beyond the role of CNP as an oocyte meiotic arrest factor, previous studies by our group and others confirmed that adding CNP to the pre-IVM system significantly improved oocyte maturation and subsequent embryo developmental potential (Richani & Gilchrist, 2022; Soto-Heras, Paramio, & Thompson, 2019; Wei et al., 2017). The synchronization of nuclear and cytoplasmic maturation is essential for oocyte quality and supporting early embryonic preimplantation development. However, the underlying molecular mechanism and whether CNP has any beneficial effect on the maternal age-induced decline in oocyte quality are incompletely understood. In the present study, we showed that CNP levels declined with age and demonstrated that CNP supplementation increased the number of antral follicles and the ovulation rate and enhanced oocyte quality and fertility. Furthermore, supplementation of CNP in pre-IVM oocyte culture medium reversed the adverse effects of age on immature oocytes, offering a potentially effective approach for assisted reproductive technologies to acquire a greater number of high-quality oocytes and improve the fertility of older women.

Many factors affect the adverse effects on the oocyte maturation process and embryonic development associated with advanced maternal age (Mikwar, MacFarlane, & Marchetti, 2020). Assisted reproductive technologies are an efficient scheme to resolve infertility as maternal fertility declines with ageing. However, the low success rate of IVM oocytes, which is especially pronounced in maternally aged oocytes, limits fertilization outcomes. Our *in vivo* results showed that CNP supplementation results in multiple improvements, including reductions in oxidative damage, spindle defects, and abnormal organelle distributions and functions, in maternally aged oocytes. Thus, we further investigated the use of CNP in the IVM system, especially in improving the quality of oocytes derived from aged mice. The results indicated that CNP-induced temporary meiotic arrest improved the maturation and fertilization rate of maternally aged oocytes and increased their subsequent embryo developmental competence. Specifically, our results confirmed an advanced role for CNP in preventing the development of mitochondrial structure abnormalities and the typical dysfunctional processes in aged oocytes. Furthermore, these data showed that CNP apparently partially rescued the antioxidant defence system impairment accompanying oocyte ageing and alleviated oxidative stress. In addition, the findings demonstrated that CNP improved cytoplasmic maturation events by maintaining normal CG, ER and Golgi apparatus distribution and function in aged oocytes.

The asynchronous nature of nuclear and cytoplasmic maturation is a major challenge in improving the quality of IVM oocytes (Coticchio et al., 2015). Age-related aberrant chromosome alignment prior to cytoplasmic maturation may result in poor oocyte quality and subsequent reduced reproductive potential (Russ, Haywood, Lane, Schoolcraft, & Katz-Jaffe, 2022). In the present study, after COCs were induced, the temporary meiotic arrest resulting from CNP treatment significantly increased the maturation rate, which may synchronize oocyte nuclear and cytoplasmic maturation. Organelle distribution is a necessary feature of oocyte cytoplasmic maturation and subsequent development. Dramatic ER reorganization (FitzHarris, Marangos, & Carroll, 2007), CG translocation to the cell cortex (Liu, Sims, Calarco, & Talbot, 2003), and the Golgi apparatus distribution and function are commonly regarded as indicators of cytoplasmic maturation (Mao, Lou, Lou, Wang, & Jin, 2014). Similarly, our results demonstrate that CNP supplementation restored cytoplasmic maturation in maternally aged oocytes by ensuring normal organelle distribution dynamics and organelle function, increasing the fertilization capacity and developmental competence of aged oocytes. It is reasonable to assume that CNP is a potential option to prevent abnormal organelle distribution and functions in oocytes that could be triggered by ubiquitous environmental endocrine disruptors, such as bisphenol A and citrinin (Pan et al., 2021; Sun et al., 2020). Cortical granules (CGs) are oocyte-specific vesicles located under the subcortex. Fusion of CGs with the oocyte plasma membrane is the most important event needed to prevent polyspermy (Miao et al., 2020). The distribution of CGs is usually regarded as one of the most important indicators of oocyte cytoplasmic maturation. The contents of

the CGs are normally discharged by exocytosis when the egg is stimulated by the fertilizing spermatozoon; this process is called the cortical reaction, and it prevents polyspermy and protects the embryo from a hostile environment during early development (Schuel, 1978).

One of the major known causes of oocyte oxidative damage and apoptosis arises from excessive ROS accumulation with ageing, especially in IVM oocytes (Combelles, Gupta, & Agarwal, 2009; Soto-Heras & Paramio, 2020). Excessive ROS accumulation occurs as a result of two processes, namely, constant generation in the mitochondria or scavenging by antioxidant defence systems, both of which involve age-related quality decreases in oocytes (Mianqun Zhang, Lu, Chen, Zhang, & Xiong, 2020). Thus, maintaining the balance between the production and scavenging of ROS could help to alleviate age-related oxidative damage and fertility decreases. Some antioxidative factor(s) within oocytes might deteriorate as the potential mother ages, compromising the ability for ROS scavenging (Schwarzer et al., 2014). GSH serves as one of the antioxidants in oocytes to combat ROS-mediated oxidative stress, which is highly correlated with oocyte developmental competence (Furnus et al., 2008). The present results suggested that CNP-induced temporary meiotic arrest increased the GSH/GSSG ratio, which is involved in the enhancement of oocyte antioxidant defence and may contribute to improving oocyte developmental competence. Consistent with previous studies (Miao et al., 2020), our findings validated that maternal ageing results in excessive accumulation of ROS and DNA damage, which severely impairs follicle development, ovulation, oocyte quality and subsequent embryo developmental potential.

Defects in chromosome separation and decondensation as well as chromosomal misalignment caused by spindle detachment are the major contributing factors responsible for the decline in oocyte quality with ageing (Chiang, Schultz, & Lampson, 2011; Eichenlaub-Ritter, Vogt, Yin, & Gosden, 2004). Oocytes require ATP for spindle formation, chromosome segregation, and polar body extrusion and fertilization processes (Arhin, Lu, Xi, & Jin, 2018; Eichenlaub-Ritter, 2013). Mitochondria are the most abundant organelles in oocytes and play an important role in ATP production via oxidative phosphorylation to phosphorylate adenosine diphosphate (Bentov, Yavorska, Esfandiari, Jurisicova, & Casper, 2011). Thus, mitochondrial function is a key indicator of oocyte quality and successful fertilization in assisted reproductive technologies (ARTs) (Mikwar et al., 2020). Mitochondrial metabolic activity and mitochondrial DNA replication dramatically decrease in oocytes with maternal age, which reduces ATP production; leads to meiotic spindle damage, chromosome misalignment and aneuploidy; and largely impairs oocyte maturation processes (Eichenlaub-Ritter, Wieczorek, Lüke, & Seidel, 2011; May-Panloup et al., 2016). We demonstrated that CNP reverses mitochondrial dysfunction induced by ageing in oocytes by analysing the mitochondrial distribution, ATP content, and mitochondrial membrane potential ($\Delta\Psi_m$).

Notably, disruption of the mitochondrial membrane potential is a potent trigger of mitophagy (Matsuda et al., 2010). Our single-cell transcriptome profiling data showed that the expression of genes related to ubiquitin-mediated proteolysis and the mitophagy pathway was considerably upregulated in aged oocytes but restored to normal levels following CNP supplementation. We also observed by TEM that CNP supplementation suppressed the accumulation of autophagic vesicles containing mitochondria. Furthermore, immunoblot analysis revealed degradation of the autophagy biomarker p62 and accumulation of LC3-II in aged oocytes, events that were markedly suppressed in CNP-treated oocytes. In general, excessive activation of mitophagy and mitochondrial damage in aged oocytes may be involved in the deterioration of oocyte quality, while CNP can ameliorate this process.

The PINK1/Parkin pathway is one of the most studied ubiquitin-dependent mitophagy processes and is crucial for the equilibrium between mitochondrial biogenesis and mitochondrial removal via selective recognition and elimination of dysfunctional mitochondria (Ateneo, 2019). In healthy mitochondria, the serine/threonine kinase PINK1 is

usually expressed at low levels, but it rapidly accumulates on damaged or aged mitochondria that exhibit loss of membrane potential (Narendra & Youle, 2011). The decrease in the mitochondrial membrane potential abolishes translocation across the outer and inner membranes and confines PINK1 in the mitochondrial matrix, stabilizing it on the mitochondrial outer membrane in a complex with the translocase TOM (De Gaetano et al., 2021). Stabilized PINK1 recruits the cytosolic E3-ubiquitin ligase Parkin from the cytosol to damaged mitochondria, an event followed by mitophagy. The effects of CNP on meiotic arrest depend on the maintenance of cAMP levels in oocytes (Meijia Zhang et al., 2010). Recent findings revealed that cAMP-dependent activation of PKA reduced the PINK1 protein level due to its rapid degradation via the proteasome and severely inhibited Parkin recruitment to depolarized mitochondria (Lobo et al., 2020). Our data confirmed that PINK1 expression was decreased in CNP-treated oocytes, thus leading to a reduction in Parkin recruitment. However, these effects were disrupted by the inhibition of PKA pathway activity, indicating that cAMP indeed mediates the ameliorating effects of CNP on aged oocyte quality. Taken together, these findings indicate that PKA-mediated inhibition of Parkin recruitment may contribute to protecting mitochondria with a low membrane potential from mitophagy in aged oocytes.

Collectively, our studies demonstrated that CNP improves the fertilization and developmental competence of maternally aged mouse oocytes by preventing age-related antioxidant defects and excessive mitophagy. Considering its known contributions as a physiological meiotic inhibitor, CNP provides an alternative to prevent maternal age-related oocyte quality defects and improve developmental competence. Although ARTs have been widely used to treat infertility, their overall success rates in women of advanced maternal age remain low. Our data may provide a new theoretical basis for the use of CNP in improving subfertility in older women or the application of clinically assisted reproduction. Out of caution, however, randomized controlled clinical trials should be conducted to further study the efficacy of CNP in women who wish to become pregnant.

Materials and methods

Animals and ethics statement

The young (6–8-week-old) and aged (58–60-week-old) C57BL-6J female mice were obtained from the Experimental Animal Center of the Xi'an Jiaotong University and housed in a temperature (20–25°C) and light-controlled environment (12 h light–12 h dark cycle) and provided with food and water ad libitum. All the experimental protocols and mice handling procedures were reviewed and approved by the Institutional Animal Care and Use Committee of the College of Veterinary Medicine, Northwest A&F University (No. 2018011212).

In vivo treatment with CNP

Aged mice (58-week-old) were intraperitoneally injected daily with CNP (120 µg/kg body weight; Cat#B5441, ApexBio) for 14 days. CNP was dissolved in PBS and diluted to appropriate concentration by physiological saline solution before injection. The mice were followed by a single injection of 5 IU pregnant mare serum gonadotropin (PMSG; Ningbo Second Hormone Factory, Ningbo, China) for 46 h to stimulate penultimate follicle maturation before collection of ovaries for histological analyses and weighting. Some mice were further injected with 5 IU human chorionic gonadotropin (hCG; Ningbo Second

Hormone Factory, Ningbo, China), and the ovulated oocytes in oviducts were monitored 16 h later to evaluate ovulation efficiency.

Measurement of CNP levels

CNP was measured in plasma by a two-site polyclonal direct ELISA kit (Biomedica Medizinprodukte, Vienna, Austria) according to the manufacturer's instructions. Collect blood samples in standardized serum separator tubes (SST), allow samples to clot for 30 minutes at room temperature and perform serum separation by centrifugation. Assay the acquired serum samples immediately. Read the optical density (OD) of all wells on a plate reader using 450 nm wavelength. Construct a standard curve from the absorbance read-outs of the standards. Obtain sample concentrations from the standard curve.

Histological analysis of ovaries

Ovaries from each group of mice were fixed in 4% paraformaldehyde (pH 7.5) overnight at 4°C, dehydrated using graded ethanol, followed by xylenes and embedded in paraffin. Paraffin-embedded ovaries were serial sectioned at a thickness of 5 µm for hematoxylin and eosin (H&E) staining. Ovaries from three mice of each group were used for the analysis.

Collection and culture of cumulus-oocyte complexes (COCs)

Female mice were stimulated by an intraperitoneal injection of 5 IU PMSG, and mice were sacrificed by cervical dislocation 24 h later. The ovaries were collected, and the well-developed Graafian follicles were punctured with 30-gauge needles to collect COCs. Only COCs with morphological integrity and a distinct germinal vesicle (GV) were cultured in basic culture medium consisted of Minimum Essential Medium (MEM)-α (Life Technologies, New York, USA) supplemented with 3mg/mL bovine serum albumin and 0.23 mM pyruvate at 37°C under an atmosphere of 5% CO₂ in air with maximum humidity.

CNP treatment and in vitro maturation

For in vitro induce temporary meiotic arrest, COCs were cultured in basic culture medium containing 10 nM of CNP. The dose of CNP for meiotic arrest in mouse oocytes in vitro was selected based on the published literatures (Meijia Zhang et al., 2011) and our preliminary reports (Wei et al., 2017). After meiotic arrest culture for 24 h, COCs were transferred to CNP-free IVM medium (containing 10 ng/mL epidermal growth factor (EGF)) to induce maturation. After incubation for 16 h, COCs were denuded of cumulus cells by treatment with 0.03% hyaluronidase to obtain MII oocytes for future experiments.

In vitro fertilization and embryo culture

Caudae epididymides from 12-week-old male C57BL-6J mice were lanced in a dish of in human tubal fluid (HTF) medium to release sperm, followed by being capacitated for 1 h (37°C under an atmosphere of 5% CO₂ in air with maximum humidity). Matured oocytes were incubated with capacitated sperm at a concentration of 4×10^5 /mL in 100 µL HTF for 6 h at 37°C, 5% CO₂. The presence of two pronuclei was scored as successful fertilization. The embryos were cultured in KSOM under mineral oil at 37 °C in 5% CO₂ and saturated humidity.

Preantral follicle isolation and culture

Ovaries were removed after the animals had been killed by cervical dislocation and preantral follicles were mechanically isolated using 26-gauge needles. Then, the preantral follicles (80–100µm diameter) that enclosed by an intact basal membrane were collected, distributed randomly and cultured individually in 96-well tissue culture plates for up to 6 days at 37°C in a humidified atmosphere of 5% CO₂ in air. The basic culture medium consisted of MEM-α supplemented with 1mg/mL BSA, 1% ITS (5µg/mL insulin, 5µg/mL transferrin, 5ng/mL selenium; Sigma), 100µg/mL sodium pyruvate and 1% penicillin/streptomycin sulfate (Sigma) in the absence (control) or presence of 100nM CNP or 10 µM H89. The dose of CNP was selected based on the published literatures (Xi et al., 2019). Half the medium was replaced with fresh medium and follicles were photographed every other day, and follicle diameter was measured using ImageJ at each time point.

Immunofluorescent Staining

Oocytes were fixed in 4% paraformaldehyde in PBS for 30 min at room temperature, and permeabilized with 0.5% Triton X-100 for 20 min, then blocked with 1% BSA in PBS for 1 h at room temperature. The oocytes were incubated with primary antibodies (Alexa Fluor® 488 Conjugate anti-α-tubulin monoclonal antibody, 1:200, Cell Signaling, Cat#35652; rabbit anti-Tom20 antibody, 1:100, Cell Signaling, Cat#sc-42406; mouse anti-Parkin antibody, 1:100, Santa Cruz, Cat#sc-32282) at 4°C overnight, and then the oocytes were extensively washed with wash buffer (0.1% Tween 20 in PBS), probed with Alexa Fluor 488 goat anti-rabbit IgG (1:200, Thermo Fisher Scientific, A21206) or Alexa Fluor 594 donkey anti-mouse IgG (1:200, Abcam, ab150108) in a dark room for 1 h at room temperature. Then oocytes were counterstained with DAPI (10 µg/mL) at room temperature for 10 min. Finally, samples were mounted on glass slides and observed viewed under the confocal microscope (Nikon A1R-si).

Mito-Tracker, ER-Tracker, and Golgi-Tracker Red Staining

Oocytes were incubated with Mito-tracker Red (1:2000, Beyotime Biotechnology, Shanghai, China), ER-Tracker Red (1:3000, Beyotime Biotechnology), Golgi-Tracker Red (1:50, Beyotime Biotechnology) in M2 medium for 30 min at 37 °C in a 5% CO₂ and saturated humidity. Then, the oocytes were counterstained with DAPI (10 µg/mL) for 5min at 37 °C in a 5% CO₂ and saturated humidity, and finally, the samples were washed 3 times with M2 medium and examined with a confocal laser-scanning microscope (Nikon A1R-si).

Mitochondrial membrane potential ($\Delta\Psi_m$) measurement

Oocyte mitochondrial membrane potential was evaluated using Mito-Probe JC-1 Assay Kit (Beyotime Institute of Biotechnology, Shanghai, China). Briefly, oocytes were incubated with 2 µM JC-1 in M2 medium for 30 min at 37 °C in a 5% CO₂ and saturated humidity, and finally, the samples were washed 3 times with M2 medium and examined with a confocal laser-scanning microscope (Nikon A1R-si). JC-1 dye exhibits a fluorescence emission of green (529 nm) and red (590 nm). Thus, the red/green fluorescence intensity ratio was measured to indicate mitochondrial depolarization. Oocytes mitochondrial membrane potential ($\Delta\Psi_m$) measurements were performed as our previous report (H. Zhang et al., 2020).

Monitoring of ROS levels in oocytes

The amount of ROS in oocytes was processed with 10 μ M oxidation sensitive fluorescent probe dichlorofluorescein (DCFH) (Beyotime Institute of Biotechnology, Shanghai, China) for 30 min at 37°C in M2 medium. Then oocytes were washed three times with M2 medium and placed on glass slides for image capture under a confocal microscope (Nikon A1R-si).

Determination of mitochondrial ROS (mitoSOX) generation by MitoSOX staining, GV oocytes were incubated in M2 media containing 5 μ M MitoSOX Red (ThermoFisher, M36008, Waltham, USA,) in humidified atmosphere for 10 min at 37°C. After washing three times in M2 media, oocytes were imaged under a confocal microscope (Nikon A1R-si).

Measurement of the GSH/GSSG ratio

The GSH/GSSG ratio was measured with a GSSG/GSH Assay Kit (Beyotime Institute of Biotechnology) according to the manufacturer's instructions. Briefly, oocytes were lysed in 40 μ L deproteinized buffer on ice for 10 min. The lysate was centrifuged at 12000 $\times$ g for 5 min at 4 °C. For GSSG measurement, the samples were incubated with GSH scavenge buffer for 60 min at 25 °C to decompose GSH. Then, the samples were transferred to the 96-well plates and the absorbance was measured with a multimode plate reader (BioTek Epoch) at 412 nm.

Measurement of the oocyte NADPH content

The oocyte NADPH contents were measured using a NADPH assay kit (Beyotime Institute of Biotechnology) according to the manufacturer's instructions. Briefly, 50 ~ 60 oocytes per group were lysed in 100 μ L NADPH extraction buffer on ice for 20 min. After the samples were centrifuged at 12000 $\times$ g for 5 min at 4 °C, the supernatants were transferred to the 96-well plates (50 μ L per well), and the absorbance was measured using a multimode plate reader (BioTek Epoch) at 450 nm. The amount of NADPH was determined using a calibration curve.

Western blot analysis

Approximately 200 oocytes were lysed in RIPA buffer (solarbio, Beijing, China) that supplemented with 1mM protease inhibitors phenylmethylsulfonyl fluoride (PMSF, solarbio, Beijing, China), on ice for 30 min. Samples were boiled at 100 °C in a metal bath for 10 min in protein loading buffer (CoWin Biosciences, Beijing, China) and equal amount proteins were separated by 10% SDS-PAGE gel, and transferred to polyvinylidene fluoride (PVDF) membranes (Millipore, Bedford, USA). After transfer, the membranes were blocked in TBST that contained 3 % BSA for 1 h at room temperature, followed by incubation with primary antibodies at 4 °C overnight (the primary antibodies were rabbit anti-GAPDH antibody, 1:2000, Cell Signaling, Cat#5174; rabbit anti-p62 antibody, 1:1000, Cell Signaling, Cat#23214; rabbit anti-Tom20 antibody, 1:1000, Cell Signaling, Cat#sc-42406; rabbit anti-LC3A/B antibody, 1:1000, Abcam, Cat#ab128025; rabbit anti-PINK1 antibody, 1:1000, Cell Signaling, Cat#6946; mouse anti-Parkin antibody, 1:1000, Santa Cruz, Cat#sc-32282). The secondary antibodies were incubated for 1 h at room temperature, and then the membrane signals were visualized by a chemiluminescent HRP substrate reagent (Bio-rad Laboratories, Hercules, CA, USA) and images were captured with Tanon5200 Imaging System (Biotanon, Shanghai, China). The band intensity was assessed with Image J software and normalized to that of GAPDH.

Transmission electron microscope (TEM)

Oocytes were prefixed with 3% glutaraldehyde, refixed with 1% osmium tetroxide, dehydrated in acetone, and embedded in Ep812 (Can EM Ltd.). Semithin sections were stained with toluidine blue for optical positioning, and ultrathin sections were made with a diamond knife and observed by a JEM-1400FLASH transmission electron microscope (JEOL) after staining with uranyl acetate and lead citrate.

Evaluation of total ATP content

The ATP content of oocytes was detected with an ATP Bioluminescence Assay Kit (Beyotime Institute of Biotechnology). The oocytes were lysed with 50 μ l of ATP lysis buffer on ice, centrifuged at $12000 \times g$ at 4°C for 5 minutes, and the supernatants were transferred to a 96-well black culture plate. Then, the samples and standards were read with a Multimode Microplate Reader (Tecan Life Sciences). Finally, the ATP level was calculated according to the standard curve.

RNA sequencing and analysis

GV-stage oocytes were collected from three young, three aged and three CNP treated aged mice. mRNA samples were collected from 5 oocytes from the same mouse of each group. The mRNA was directly reverse-transcribed through oligo dT. The reverse-transcribed cDNA was amplified, and the cDNA was cut by Tn5 transposase digestion, and linkers were added to obtain the required sequencing library. The constructed library was entered into the sequencing program after passing through an Agilent 2100 Bioanalyzer and RT-PCR quality control. The PE100 sequencing strategy was used to assess gene expression changes at the transcription level. Dr. Tom was used for difference analysis, GO analysis, KEGG analysis, and other analyses.

Reverse transcriptase quantitative PCR (RT-qPCR) analysis

Total RNA from oocytes was extracted using MiniBEST Universal RNA Extraction Kit (TaKaRa, Dalian, China), and reverse transcribed to synthesize cDNA using a PrimeScript RT Master Mix reverse transcription kit (TaKaRa) according to the manufacturer's instructions. RT-qPCR quantitation of mRNAs was performed using TB Green™ Premix Ex Taq™ II (TaKaRa) with Applied Biosystems StepOnePlus Real-Time PCR System (Thermo Fisher Scientific, Massachusetts, USA) using the following parameters: 95 °C for 1 min, followed by 40 cycles at 95 °C for 5 s and 60 °C for 34 s. The PCR primers used in this study were shown in Table S1. Transcript levels were normalized to those of the housekeeping gene *Gapdh*. The CT value was used to calculate the fold change using the $2^{-\Delta\Delta C_t}$ method. Each experiment was repeated independently at least thrice.

Statistical analysis

Statistical analyses were performed using GraphPad Prism 8.00 software (GraphPad, CA, United States). Data were reported as means $\pm$ SEM. Results of statistically significant differences were denoted by asterisk. ($P < 0.05$ denoted by *, $P < 0.01$ denoted by **, $P < 0.001$ denoted by ***, and $P < 0.0001$ denoted by ****).

Ethics approval

The experimental protocols and mice handling procedures were reviewed and approved by the Institutional Animal Care and Use Committee of the College of Veterinary Medicine, Northwest A&F University (No. 2018011212).

Conflicts of interest

All of the contributing authors declared no conflicts of interest. All authors read and approved the final manuscript.

Data availability statement

All data is included in the manuscript and supporting information.

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Joint Public Review:

It has been shown previously that maternal aging in mice is associated with an increase in accumulation of damaged mitochondria and activation of parkin-mediated autophagy (see DOI: 10.1080/15548627.2021.1946739). It has also been shown that C-natriuretic peptide (CNP) regulates oocyte meiotic arrest and that its use during in vitro oocyte maturation can improve parameters associated with decreased oocyte quality. Here the authors tested whether use of CNP treatment in vivo could improve oocyte quality and fertility of aged mice, for which they provided convincing evidence. They also attempted to determine how CNP improves oocyte developmental competence. They showed a correlation between CNP use in vivo and the appearance (and some functional qualities) of cytoplasmic organelles more closely approximating those of oocytes from young mice. However, this correlation could not be interpreted to imply causation. Additional experiments performed using CNP during in vitro maturation were not properly controlled and so are not possible to interpret.

A strength of the manuscript is that the authors use an in vivo treatment to improve oocyte quality rather than just using CNP during oocyte maturation in vitro as has been done previously. This strategy provides more potential for improving oocyte quality - over the course of oocyte growth and maturation - rather than just the final few hours of maturation alone. This strategy also has the potential to be translated into a more generally useful clinical therapeutic method that using CNP during in vitro maturation. However, it is difficult to glean information regarding how CNP might have its effects in vivo. A range of models are used in the manuscript with a mix of in vivo studies with in vitro experiments, which results in some disconnect between systemic CNP and its reported intrafollicular action as well as in the short-term versus longer-term actions of CNP on oocyte quality. Specifically, CNP was shown to be reduced in the plasma of aged mice, but this was not shown in the granulosa cells, which are the reported source of CNP that acts on oocytes. Whether the ovarian source of CNP is reduced in aged females was not demonstrated, and CNP is not known to act on oocytes through an endocrine effect. In vivo treatments with CNP by i.p. injection were performed, but the dose (120 ug/kg) and time (14 days) of treatment were not validated by any prior experiments to give them physiological relevance.

Weaknesses:

1. There are errors in the manuscript writing that make the Results difficult to follow. Reference to the Figures in the Results section does not match what is shown in the Figure panels. For example, the Results text reports differences in CNP levels in aged and young mice shown in Figure 1C, but the relevant panel is actually shown in Figure 1F. Other Figures have the same problem.
2. The Results section is not always clear regarding what CNP treatment was done - in vivo injections or in vitro maturation. For example, what is the difference, if any, between Figures 2C-D and Figures S2A-B?
3. Immature oocytes from aged females (~1 year) were treated with a two-step culture system with a pre-IVM step with CNP. Controls included oocytes from young (6-8 weeks) females or oocytes from aged females treated by conventional IVM. The description of these methods suggests that control oocytes did not receive an equivalent pre-IVM culture, hence the relevance of comparisons of CNP-treated versus control oocyte is questionable. It was observed that aged oocytes pre-cultured in CNP improved polar body extrusion rates and meiotic spindle morphology compared to oocytes in conventional IVM, as has been well established. The description of statistical methods does not make clear whether the PBE rate in CNP-treated old oocytes remained significantly lower than young controls.
4. The main effect of the CNP 2-week treatment appears to be increasing the number of follicles that grow into secondary and antral stages, but there is no attempt made to discover the mechanism by which this occurs and therefore to understand why there might be an increase in the number of ovulated eggs, quality of the eggs, and litter size. It is also not clear how an intraperitoneal injection can guarantee its effectiveness because the half-life of CNP is very short, only a few minutes.
5. Meiotic spindle morphology, as well as a number of putative markers of cytoplasmic maturation are also suggested to be improved after pre-culture with CNP. In each case a subjective interpretation of "normal" morphology of these markers is derived from observations of the young controls and the proportions of oocytes with normal or abnormal appearance is evaluated. However, parameters that define abnormal patterns of these markers appear to be subjective judgements, and whether these morphological patterns can be mechanistically attributed to the differences in developmental potential cannot be concluded.
6. In addition to the localization patterns of mitochondria, the mitochondrial membrane potential, oocyte ATP content and ROS levels were assessed through more objective quantitative methods. These are well known to be defective in oocytes of aged females and CNP treatment improved these measures. Mitochondrial dysfunction is the most obvious link between oocyte apoptosis, autophagy, cytoplasmic organelle miss-localization and aberrant spindle morphology. Among the most intriguing results is the finding that CNP mediated a cAMP-dependent protein kinase (PKA) dependent reduction in mitochondrial autophagy mediators PINK and Parkin and reduced the recruitment of Parkin to mitochondria in oocytes. However, it may not be possible to directly link this observation to the improvements in IVM oocyte quality, since PINK/Parkin assessments were performed in oocytes from cultured follicles treated with CNP for 6 days.
7. The gold standard assay for oocyte quality is embryo transfer and live birth. The authors assessed the impact of maturing oocytes in vitro in the presence of CNP on oocyte quality by less robust assays (e.g., preimplantation embryo development in vitro), so the impact on oocyte quality is less certain.
8. The terminology used to describe many of the Results exaggerates the findings. For example, the authors claim that many of their immunofluorescent markers of the various organelles have a pattern that is "restored" by CNP. However, in most cases the pattern is "improved" toward the control condition but is not fully restored.
9. The numbers of embryos should have been corrected for the number of eggs fertilized as a starting point so that the percentage that developed to each stage could be expressed as a percentage of successfully fertilized eggs rather than overall percentages. As currently

shown in the Figures and described in the Legend, there is no information regarding what the percentage on the y-axis means. For example, does Figure 4B show the number of 2C embryos divided by the number of eggs inseminated? Or is it divided by the number of successfully fertilized eggs, and if so, how was that assessed?

10. When fewer eggs are fertilized, the numbers of embryos per group are lower and so the impact of culturing multiple embryos together is lost. As a result, it is possible that culture conditions rather than oocyte quality drove the differences in the numbers of embryos that achieved each stage of development.

11. Not all claims in the Discussion are supported by the evidence provided. For example, "In addition, the findings demonstrated that CNP improved cytoplasmic maturation events by maintaining normal CG, ER and Golgi apparatus distribution and function in aged oocytes" but it was never demonstrated that the altered distribution had any functional impact.

12. Incompleteness and errors in the Methods section reduce confidence in many of the results reported.

13. The methods used for Statistical Analysis are never explained in either the Methods or the Figure legends. It is unclear whether appropriate analyses were done, and it is frequently unclear what was the sample size and how many times a particular experiment was repeated. These weaknesses detract from confidence in the data.